

The Shift From Imaginary Friends to Synthetic Beings in Current Childhoods

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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Introduction

As Artificial Intelligence (AI) has begun its push towards consumers, younger audiences have gotten hold of the emerging technologies. Access to technology among children has been observed to cause many adverse effects, including loneliness, depression, and, in extreme cases, suicide (Clemente-Suárez et al., 2024). Yet AI has also been observed to increase cognitive skills among children in educational environments (Abrar et al., 2025). Artificial Intelligence poses numerous new dangers and concerns for school-aged children; however, it can enhance various cognitive skills when used in conjunction with strict precautions. This paper asks: What effects does AI have on children in home environments, and how does this help/hinder their social and cognitive development? I analyze various sources related to children and past technologies, as well as children and AI. Technological momentum will be used to analyze the developmental stages of AI technologies and pinpoint how developers can use these stages to their advantage to prevent harm to children, before the technology becomes too complex to modify (Nye, 2007, pp. 52-56). Technological momentum describes the progression of technologies, typically infrastructure similar to electrical grids and road systems, and how these systems go from impressionable and flexible to set in stone over time (Nye, 2007, pp. 52-56).

AI is harmful for children when used in home environments where children have more technological freedom, as compared to the close monitoring and restrictions in educational environments. Developers need to consider both environments when developing AI as well as child audiences as a whole, as children are subject to harm and danger when using AI in its current state due to their developing and vulnerable minds. To support these claims, this paper consists of an analysis of the effect of past technologies (iPads and social media) on children, device and AI dependencies among children, parasocial relationships and interactions between

children and AI, and finally, the positive impacts of AI within strict confines and adult supervision.

Previous Technological Advances' Impact on Children's Well-being

New digital technology products have presented many concerns for users, especially school-aged users. Examples of strong and highly influential digital technology releases include: iPads/tablets, social media, and mobile devices. These technologies have been popularized much longer than artificial intelligence, offering a glimpse into the future of children and AI. Clemente-Suárez et al. provide many insights into the harm of technology on children, presenting direct concerns for child health and even emotional development. An increase in screentime for children is directly correlated with increased amounts of depression and loneliness (Clemente-Suárez et al., 2024, p. 20). Screentime also causes higher rates of physical inactivity among children as well as decreased fine motor skills (Clemente-Suárez et al., 2024, p. 14). Additionally, “the negative consequences of excessive or inappropriate digital device usage include reduced attention spans and sleep disruptions” (p. 22). This quote further emphasizes the impact of digital technology on children, as attention spans and sleep are integral for an individual to succeed day-to-day, as well as to effectively socialize and create bonds with peers.

Another good example of interactions between children and digital technology is Sherry Turkle's book *Alone Together: Why We Expect More From Technology and Less From Each Other* (2017), which details the relationships between various individuals and technology. Particularly, the thirteenth chapter, “Anxiety,” focuses on teens and their reflections on the impact technology has had on them. The chapter focuses on the introduction of mobile cell phones and social platforms, such as Facebook. Julia, a 16-year-old, details her attachment to her

cell phone and how she feels “safe” with it, particularly after the 9/11 attacks (p. 243). She goes on to detail how she has formed an unhealthy attachment to her phone and the convenience of texting, so much so that being without her phone or being unable to read text messages immediately makes her “antsy” (p. 245). Another 16-year-old, Hannah, details her experience with online chats and how they began light-hearted and helped her build confidence, but quickly became a source of anxiety in her daily life (p. 248). Hannah had begun chatting online to help ease some of her anxiety about talking to boys when she met Ian online, and he became her emotional partner (p.249). This interaction raises concerns regarding the validity surrounding the claimed ages of the individuals, as well as the emotional pressure of a digital relationship on a minor. As time went on and Hannah became more active online, she began to feel pressured from her online peers to be “present” in their online community, resulting in her committing nearly 20 hours of her week to maintaining her online social status (p. 250). She became so isolated from her in-person friends that she believed her online friends were all she had, and she grew anxious to respond to them because they’d get mad if she didn’t respond promptly. The chapter continues with several more cases, all of which set a clear precedent for the possible effects of AI, particularly AI companions and the pressure and attachment children may feel from them, similar to Hannah and Ian, as well as her other online friends.

Some may argue that new digital technology has had as many positive impacts as it has negative impacts on children. It is undeniable that these new releases can positively impact students, with the potential to improve both empathy and collaboration (Clemente-Suárez et al., 2024, p. 22). However, when thinking of the positive and negative impacts of digital technology, it is imperative to consider the lingering effects of these impacts. For example, increased depression and anxiety have the potential to lead to suicidal thoughts and tendencies, a possibly

irreversible path that can cause fatal harm. These impacts are important to think about when considering the interactions between children and AI. Mansfield et al. (2025) put it into frame well: “If history is any guide, research and evidence-based policy will lag behind these new interactions.”(p. 202). There is a strong stress on the involvement of regulations and AI in an attempt to mitigate potential harm from these resources and guide legal guardians when making their decisions. The American Psychological Association even released a comprehensive guide for AI use and exposure for young US citizens, warning of potential harms of artificial intelligence and relating the new surge in AI to past digital technologies, similar to Mansfield et al. (2025). Digital technology introduces cyberbullying, inappropriate content, and binge-watching, all of which are harmful to children. It has also encouraged the social isolation of individuals, as they can gain a sense of the same fulfillment of social situations, but instead through a device. This creates divides with peers and further isolates children, encouraging them to turn deeper into their devices as described by Hannah in Turkle’s book (2017), (Clemente-Suárez et al., 2024, p. 20). Overall, previous digital technology releases present a dangerous precedent that strongly suggests artificial intelligence will harm children and further hinder social development, along with increasing mental illness among youth.

Children Can Become Dependent on Devices and Artificial Intelligence

A major concern of the rise of artificial intelligence among children is the impact it may have on their cognitive development, including problem-solving, critical thinking, and other essential skills in today's society. When children begin to rely heavily on AI, independent skills necessary as foundational building blocks, such as critical thinking and other decision-making skills, take a back seat in terms of importance (Abrar et al., 2025, p. 3). AI makes seemingly small skills seem unimportant and easy to delegate to an AI resource, since it will do it much

faster. One example of this is summarizing articles and other resources (Clemente-Suárez et al., 2024, p. 12). Using AI to engage with informative media encourages shallow processing of information and “can hinder memory consolidation and the ability to form long-term memories” (Clemente-Suárez et al., 2024, p. 12). If these skills are stunted at a young age, it can be detrimental to critical thinking and a child's autonomy as they grow older, including how they consume media, surf the web, converse on sensitive and political topics, and fact-check information they see online.

AI has become notorious for its ‘hallucinations,’ meaning that occasionally it will make up information that isn’t true or is dangerous, such as putting glue on pizza (McMahon, 2024). This is dangerous for individuals who lack essential cognitive skills. The APA warns that “AI-generated summaries may not always be accurate, and students must not become overly dependent on AI, which could impede developing their own knowledge and skills.” (American Psychological Association, 2025, p. 9). Alternatively, these hallucinations can promote cognitive development when children work to disprove the misinformation and assess the validity of the claims made by AI resources (Xu, 2025, p. 12). Overall, when hallucinations are encountered in a controlled manner, and a child knows not to immediately trust information given from LLMs (Large Language Models), it can help them think critically about the validity of the source. However, there is a high likelihood that outside of an educational environment, children become more inclined to believe an LLM off the bat, instead of questioning it, out of ease.

AI is good at thinking similarly to humans, allowing it to get pesky tasks and to-dos out of the way, which many could argue gives individuals more time to focus on pressing tasks, improving efficiency. This is true, AI can improve efficiency for adults, but it can harmfully hinder children. It can stunt their cognitive development, setting them up for difficulties

integrating into society as they grow older. AI is still in the early stages of development and has not yet become set in stone, as per the technological momentum framework (Nye, 2007, pp. 52-56). AI in a developmental state is similar to an electrical grid that is still being built and tweaked. However, once AI reaches a maintenance state, it will be significantly more difficult to modify, similar to a fully deployed electrical grid in an urban area. AI is still currently being developed and modified daily. This means that there are still many kinks that must be ironed out in these programs, including safeguards to prevent users from becoming overly reliant on artificial intelligence. For example, the recent ChatGPT launch, where OpenAI made ChatGPT less agreeable in an effort to emotionally detach users from the resource (Freedman, 2025, p. 1). Alterations like this are still simple to add as AI systems are not yet at a complexity where additions similar to ChatGPT's hinder their operational integrity. It is the responsibility of the developers behind AI systems to consider the possible outcomes of their systems before launching them, to avoid harm to individuals and difficulty integrating future requirements.

Parasocial Relationships and Interactions with Artificial Intelligence

AI can act in a very human-like manner, giving the illusion of interacting with another person for children, when in reality it is just algorithms and servers on the other side. This is dangerous because it places AI in an influential position in a child's life, rendering its responses more valuable and important to the child. This form of interaction aligns with the patterns of parasocial relationships and interactions. Gleason et al. (2017, p. 1) define parasocial relationships and interactions as “symbolic, one-sided social ties that individuals imagine with media figures and celebrities.” These types of interactions were first observed in the performing arts, where adult audiences interacted directly with performers (Hoffman et al., 2021, p. 607). Parasocial interactions (PSIs) reflect a give-and-take between the individual and performer,

while parasocial relationships (PSRs) reflect a seemingly face-to-face relationship for the observing individual. A case study of typical PSRs and PSIs seen in children is from Gleason et al. (2017), who performed a qualitative analysis on PSRs and interactions between 151 adolescents and famous individuals they are fond of. Nearly 58% of the 151 children described their connection to their chosen individuals as a relationship. Gleason et al. found that PSRs play a role in a child's identity formation. How exactly these PSRs impact a child's identity varies heavily depending on how the child imagines their relationship with their celebrity, leading to a variety of impacts on children. Their study highlights the deep impact of traditional PSRs and PSIs on children and how a child's perception of those connections can influence their self-image and perception of peers.

With the development of entertainment media and new technologies, these social ties have extended beyond media figures and celebrities. Conversational agents (CAs) are voice assistants that utilize artificial intelligence to mimic “human speech and human interaction patterns” (Hoffman et al., 2021, p. 608). Hoffman et al. analyzed the use of CAs in homes with adolescents in various ways, including both how children view and interact with the agents. Parents reported that their children engage regularly with the CAs, similarly to how they engage with media characters. These interactions included notably concerning activities, such as children personifying CAs, forming attachments to CAs, and viewing CAs as alive. Children asked many questions to these agents, including both scientific (34% of questions) and personal questions about the CA's life (15% of questions). While not all children engaged in PSRs with CAs, 89% of parents reported weekly PSIs between their child and the CA in their household. Children often perceived these CAs as friendly and trustworthy. Additionally, younger children more often perceived the CAs as personified; however, age did not affect an adolescent's

attachment to the CA. Hoffman et al.'s (2021) findings shine a light on how easily technology can become standard in a child's life, as well as the power PSRs and PSIs have on both a child's development and daily life.

Traditional PSRs and PSIs have been successfully used to foster positive social, emotional, and cognitive development for children. This is because they do not feature reciprocity, which provides children with a safe space to experiment with various ways of living (Gleason et al., 2017). Artificial intelligence does not follow the same style; CAs and LLMs are programmed to provide unique personalized responses to inputs provided by the user. While young children may use these resources for trivial purposes, such as asking "Alexa, what is the weather today?", these resources serve as a gateway to their use of the internet, setting essential and unmonitored foundational skills regarding tech literacy (Hoffman et al., 2021). Additionally, as children get older, they've been observed using "AI as a companion or emotional disclosure" (Xu, 2025, p. 8). Overall, when children engage in PSIs and form PSRs with AI, it places them in a vulnerable position where AI is used in a manner it is not designed for, creating the possibility of misleading information, harmful responses, and even emotional relationships.

As children engage in PSRs and PSIs with AI, due to its responsive nature there have been several cases of both suicide and sexual abuse from AI chatbots. A prominent example of an AI program that has harmed children in irreversible ways is Character.AI. Character.AI is an AI program designed to mimic popular media characters and enable users to engage in chats with them (Young, 2025). Character.AI is under extreme pressure as several young teens have committed suicide after chatting with their AI system. A 14-year-old, Sewell Setzer III from the state of Florida, committed suicide in March 2024 after several conversations within a Character.AI chat where the bot posed as a popular Game of Thrones character (Allyn, 2024).

Sewell had begun to isolate himself from his family and friends due to the chats, which consisted of emotional and sexual abuse from the bot. When he confided his suicidal thoughts to the bot, it responded, encouraging him to “come home” to it (Duffy, 2026). He then committed suicide. Juliana Peralta, a 13-year-old from Colorado, committed suicide in November 2023, just a few months before Sewell (Young, 2025). Juliana’s family was one of two families in Colorado suing Character.AI at the time for misconduct concerning minors, Juliana’s for her wrongful death and sexual abuse, and the other family for sexual abuse of their child within Character.AI chats. Juliana had begun isolating herself and confiding her suicidal thoughts to a chatbot “Hero” before ultimately committing suicide. The parallels of these cases are uncanny. Two children committed suicide within months of one another in a manner that should never have been an option for them. Juliana and Sewell are two cases out of several children sexually abused by Character.AI bots, many of whom fell victim to suicide. Had Character.AI developers considered the possible conversations that their bots would engage in and taken the proper precautions to shut down suicidal ideation and properly alert authorities, Juliana and Sewell, as well as the many other victims, may still be with us today.

With Supervision and Precautions, AI Increases Fundamental Cognitive Skills

Artificial intelligence, while dangerous, has also proven under certain circumstances to benefit fundamental cognitive skills in children. AI tutors have shown great success in improving problem-solving skills among children aged 7-11 years old (Abrar et al., 2025). This is because at this age, children rely on concrete previous experiences and problems they solve. AI can also tailor to different experiences of students quickly, giving them material catered to their prior knowledge and comfort levels, which further supports individual cognitive development (Abrar et al., 2025). Even the American Psychological Association (2025) recognizes the abilities of AI

through personalized feedback, explicit scaffolding, and advanced questioning techniques. Additionally, and possibly most importantly, AI makes learning more accessible for a large number of children, whether it's for those with mental or physical disabilities around 15% of school-aged children in the US benefit from special education resources (Schaeffer, 2023). These contexts consist of heavy supervision and constant monitoring of the children by well-educated professionals. Once a student is removed from this context, AI presents itself in a more malleable and dangerous manner, as the protective social scaffolding surrounding the student is removed.

Many professionals have reported the benefits of AI support for children with disabilities. Dr. LeMoine, who works directly with children with mental disabilities, writes about the positive impacts of AI on children with neurological disorders, stating that the ability to create personalized, adaptive learning environments has shown great potential in her line of work (2024, p. 3). LeMoine details four AI implementations that she's seen have positive impacts for children with disabilities. The first is speech assistants that aid children with speech impediments by offering consistent and non-judgmental environments where they can practice speaking and develop their skills. The second implementation is custom learning environments with different features for each student, such as visual and auditory stimulation for children with autism. Third is wearable technology that uses AI methods and algorithms to analyze a child's behavioral patterns in an effort to predict developing conditions, such as meltdowns and sensory overloads, allowing instructors to intervene promptly. The final implementation she mentions is AI-enhanced alternative communication methods for non-verbal children, which includes predictive methods to assist children in quicker and more efficient communication. These are all examples of the positive impacts artificial intelligence has had in an educational context. Overall,

AI can help children with disabilities develop cognitively when used in a specific manner and monitored closely, to avoid the adverse effects mentioned previously in this paper.

AI developers will begin to move from a heavily developmental state to more of a maintenance state with less emphasis on large updates and revisions. Technological momentum highlights the difficulty of altering a technology once it has been fully developed. Similar to a major city's electrical grid, which is near impossible to make significant changes and improvements to, AI will inevitably reach a state where it is not easily adjustable. AI is currently undergoing rapid development across multiple companies, making current models moldable and easy to alter to include things like less-agreeability and safeguards for child users. When developing artificial intelligence, it is becoming clear that engineers are not considering all users and possibilities of the models they are pushing. From the intense pressures of the workplace to the intended purpose of AI, it can become easy to forget about individuals who are in a susceptible position to misinformation and overreliance on these technologies. Developers must begin to consider the dangerous possibilities their technologies introduce, especially to individuals who cannot recognize right from wrong when communicating with CAs and LLMs, before AI gains tech momentum and thereby becomes harder to modify.

Conclusion

In conclusion, children's use of AI should be restricted to environments where they can be closely monitored and guided on proper use of these powerful tools. In this paper, it was discussed that social and cognitive growth can be stunted by irresponsible use of these programs. Children should not have unrestricted and unsupervised access to this new technology. Developers must be held accountable while AI is still in a developmental stage, forcing them to

protect vulnerable users. This paper has detailed the known short-term effects of AI use and the longer-term effects of preceding technologies. The scope of this paper is limited to the short-term effects of AI use on children, as insufficient time has passed since the rise of LLMs and related AI models to determine their long-term effects. Even with this barrier, it is evident that AI has the power to change a child's life forever. This power demands research into the long-term effects on children's social and cognitive development, as well as on the development of home environments to either promote safe and beneficial use or entirely block AI use in the household.

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